



Academia Internacional David
Robotics and Technology
Arduino Freenove Prototype

8th Grade - 2nd Trimester - Final Project Packet and Rubric

Name: _____

Date: _____

Teacher: Porfirio Rios Group: 8th _____

Score: ____ / 90 pts

Project Overview

Students design, build, test, debug, and demonstrate an Arduino/Freenove prototype where one input controls at least one output.

Required Evidence

- Arduino project design plan with purpose, role, input, output, diagram, behavior, tests, and materials.
- Prototype circuit evidence and starter/final sketch evidence.
- Testing log with at least three conditions.
- Debugging note with problem, likely cause, and solution.
- Final diagram, code logic explanation, demonstration, and reflection.

Schedule and Evidence Expectations

- Week 10: build first circuit version and program input/output behavior.
- Week 11: debug, improve, document, and collect final testing evidence.
- Week 12: demonstrate prototype and submit evidence.

Presentation or Demonstration Requirements

- Demonstrate the prototype or provide recorded/photo evidence.
- Identify the input, process, output, one code decision, one test result, and one improvement.

Rubric

Criterion	15-13 pts	12-9 pts	8-5 pts	4-0 pts
Design plan	Purpose, input, output, diagram, behavior, tests, and materials are complete.	Most plan parts are complete.	Plan is partial or unclear.	Plan is missing.
Circuit and build quality	Circuit is safe, labeled, and works reliably.	Circuit works with minor support.	Circuit partly works or labels are incomplete.	Circuit cannot be checked.
Code and input/output logic	Code clearly connects input to output with correct logic.	Code mostly controls the output.	Code is incomplete or only partly connected.	Code evidence is missing.

Testing and debugging	Three conditions, issue evidence, fix, and improvement are clear.	Some testing and debugging are shown.	Testing is limited or unclear.	No meaningful testing evidence.
Documentation and explanation	Diagram, code explanation, photos/screenshots, and reflection are complete.	Most documentation is present.	Documentation has gaps.	Documentation is mostly missing.
Demo and responsibility	Clear demo, safe material use, cleanup, and timely submission.	Mostly responsible work and demo.	Demo or responsibility evidence is incomplete.	Demo/submission is missing.